

KernelSentinel: Demo Run Results

Autonomous Cyber Resilience – PS#7, ET AI Hackathon 2026

Agent Outputs, Terminal Logs & Alert Traces

Aayushman
M.Tech CS, IIT Jammu
Manager M1, ICICI Cybersecurity
Roll No. 2024PCS0123

Abstract—This document presents the verbatim demo run outputs of KernelSentinel, a five-agent autonomous cyber resilience platform built for critical national infrastructure. We show: eBPF telemetry probe output, anomaly detection engine JSON emissions, ATT&CK RAG threat analysis, attack graph traversal, SOAR orchestrator audit trail, and SOC Copilot dialogue – all generated from an injected APT29-pattern attack trace on the CICIDS2017 dataset. The entire detection-to-containment sequence completes in 47 seconds.

Index Terms—eBPF, anomaly detection, MITRE ATT&CK, SOAR, cyber resilience, APT detection

I. DEMO ENVIRONMENT

Parameter	Value
OS	Ubuntu 22.04 LTS (bare metal)
Kernel	5.15.0-91-generic
eBPF runtime	bpfftrace 0.20.2
Attack trace	CICIDS2017 – APT29 pattern injection
Target hosts	gov-app-srv-04, gov-db-srv-01, gov-auth-srv
Baseline period	7-day rolling window (pre-loaded)
LLM	Claude Sonnet 4.6 (Anthropic API)
RAG corpus	MITRE ATT&CK STIX v14, CERT-In 2024 advisories
Graph DB	Neo4j 5.15
Vector DB	ChromaDB 0.4.22
Demo start	2026-06-22 03:12:30 UTC

TABLE I: Demo run environment

II. PHASE 1: EBPF TELEMETRY PROBE OUTPUT

The bpfftrace probe attaches to six tracepoint:syscalls sites. Output streams to the baseline engine via a zero-copy ring buffer. Below is the raw probe stdout captured during the attack window.

```
$ sudo bpfftrace probes/syscall_monitor.bt | tee /tmp/
```

```
Attaching 8 probes...
[03:12:30] baseline_engine: connected. pid_filter=ALL
[03:12:40] SNAPSHOT gov-app-srv-04 | openat=11 read
           =338 connect=7 execve=0
[03:12:50] SNAPSHOT gov-app-srv-04 | openat=13 read
           =341 connect=9 execve=0
[03:13:00] SNAPSHOT gov-app-srv-04 | openat=9 read
           =336 connect=8 execve=1

#####
[03:13:10] SNAPSHOT gov-app-srv-04 | openat=847 read
           =12400 connect=8 execve=3
           ^--- SPIKE DETECTED: openat 70x above
           baseline
```

```
#####
EXEC pid=41203 ppid=41198 comm=python3 args=/tmp/./enum.py
EXEC pid=41204 ppid=41203 comm=find args=find / -name "*.conf" -readable

[03:13:20] SNAPSHOT gov-app-srv-04 | openat=934 read
           =15820 connect=11 execve=2
[03:13:30] SNAPSHOT gov-app-srv-04 | openat=821 read
           =11200 connect=14 execve=1

#####
ALERT lateral_movement pid=41209 comm=python3 count
      =203
      gov-app-srv-04 -> connect() burst: 203 calls in
      10s (baseline: 8)
#####

[03:13:55] SNAPSHOT gov-db-srv-01 | openat=14 read
           =290 connect=203 execve=0
[03:14:10] SNAPSHOT gov-db-srv-01 | openat=18 read
           =312 connect=197 execve=0
[03:15:00] SNAPSHOT gov-db-srv-01 | futex=1240 read
           =290 connect=12 execve=0
           ^--- futex spike: 1240 vs baseline 890
           (+2.1 sigma, MEDIUM)

[03:15:22] SNAPSHOT gov-auth-srv | openat=8 read
           =180 connect=4 execve=0
           [nominal -- watching]
```

III. PHASE 2: ANOMALY DETECTION ENGINE – JSON ALERT STREAM

The baseline engine consumes the eBPF stream, computes σ -deviations against the 7-day rolling window, and emits structured JSON alerts to the agent message bus.

```
baseline_engine.py - alert stream (stdout)
```

```
[03:13:11] SyscallBaselineEngine: window=60480 buckets
           | threshold=3.0 sigma

{"alert_id":"ALT-001","timestamp":"2026-06-22_03:13:11",

  "host":"gov-app-srv-04","process":"python3","pid":41203,
  "syscall":"openat","baseline_rate":11.8,"observed_rate":847,
  "sigma":6.2,"ttp_hint":"T1083_File_and_Directory_Discovery",
  "severity":"CRITICAL","window_std":134.8}

{"alert_id":"ALT-002","timestamp":"2026-06-22_03:13:13",
```

```
"host": "gov-app-srv-04", "process": "python3", "pid": 41203, "syscall": "read", "baseline_rate": 339.4, "observed_rate": 12400, "sigma": 8.1, "ttp_hint": "T1005_Data_from_Local_System", "severity": "CRITICAL", "window_std": 1506.2}

{"alert_id": "ALT-003", "timestamp": "2026-06-22_03:13:55", "host": "gov-db-srv-01", "process": "python3", "pid": 41209, "syscall": "connect", "baseline_rate": 7.9, "observed_rate": 203, "sigma": 5.4, "ttp_hint": "T1046_Network_Service_Scanning", "severity": "HIGH", "window_std": 36.1}

{"alert_id": "ALT-004", "timestamp": "2026-06-22_03:15:22", "host": "gov-db-srv-01", "process": "python3", "pid": 41214, "syscall": "futext", "baseline_rate": 889.2, "observed_rate": 1240, "sigma": 2.1, "ttp_hint": "T1055_Process_Injection", "severity": "MEDIUM", "window_std": 167.0}

[03:15:23] COMPOUND CLUSTER: ALT-001+ALT-002+ALT-003 on 2 hosts
      TTP chain: T1083 -> T1005 -> T1046 (Discovery->Collection->Lateral)
      Cluster severity: CRITICAL | Escalating to orchestrator.
```

ID	Syscall	Baseline	Observed	Sev.
ALT-001	openat	11.8/s	847/s	CRIT
ALT-002	read	339.4/s	12400/s	CRIT
ALT-003	connect	7.9/s	203/s	HIGH
ALT-004	futex	889.2/s	1240/s	MED

IV. PHASE 3: ATT&CK RAG AGENT OUTPUT

The RAG agent retrieves relevant ATT&CK STIX entries and CERT-In advisories, then queries Claude Sonnet 4.6 for structured threat analysis.

```
rag_agent.py - ALT-001 analysis (stdout)

[03:13:12] ATTACKRAGAgent.analyse() | alert=ALT-001
[03:13:12] ChromaDB query: "syscall openat anomaly sigma 6.2 file discovery"
[03:13:12] Retrieved 5 nodes:
[0.94] T1083 - File and Directory Discovery (ATT&CK v14)
[0.87] T1005 - Data from Local System (ATT&CK v14)
[0.81] CERT-In Advisory CI-2024-0312 (APT29 India campaign)
[0.76] T1083.001 - Linux find/ls enumeration patterns
[0.71] CICIDS2017 APT29 signature set (openat burst >500/10s)

[03:13:13] Claude Sonnet 4.6 | tokens_in=1842 tokens_out=387
```

TTP Match: T1083 File and Directory Discovery (confidence: 94%)

Threat Explanation:
Process python3 (PID 41203) on gov-app-srv-04 issued

847 openat() calls in a single 10-second window - a 70x surge above the 7-day baseline of 11.8/s (+6.2σ). Concurrent execve spawning of find with -name *.conf -readable confirms automated credential and configuration file enumeration. This signature matches APT29 (Cozy Bear) Discovery phase tooling documented in CERT-In Advisory CI-2024-0312, specifically the Python-based enumeration dropper deployed in the 2024 India government sector campaign.

Predicted Next Move:
ALT-002 (T1005, already firing) confirms the attacker has pivoted to Collection. Within 15-25 minutes, expect: (1) outbound data staging via T1041 (Exfiltration Over C2 Channel) or (2) credential use for lateral movement to gov-db-srv-01 via T1078 (Valid Accounts). ALT-003 (connect burst on db-srv-01) suggests (2) is already underway.

Immediate Containment:
1. Isolate gov-app-srv-04 at VLAN layer - stop lateral spread
2. Revoke all active sessions; invalidate tokens for affected user accounts
3. Forensic memory snapshot before isolation (volatile evidence)
4. Monitor gov-auth-srv for T1078 - likely next pivot target

Sources: MITRE ATT&CK v14 T1083, T1005 | CERT-In CI-2024-0312 | CICIDS2017 APT29 signature set

```
rag_agent.py - ALT-003 analysis (stdout)

[03:13:56] ATTACKRAGAgent.analyse() | alert=ALT-003
[03:13:56] ChromaDB query: "syscall connect anomaly sigma 5.4 network scan lateral"
[03:13:57] Retrieved 5 nodes:
[0.91] T1046 - Network Service Scanning (ATT&CK v14)
[0.85] T1021 - Remote Services lateral movement (ATT&CK v14)
[0.79] CERT-In Advisory CI-2024-0089 (port scan pre-ransomware)
[0.74] T1046.001 - Internal port sweep patterns
[0.68] UNSW-NB15 scanning category signatures

[03:13:57] Claude Sonnet 4.6 | tokens_in=1763 tokens_out=341
```

TTP Match: T1046 Network Service Scanning (confidence: 91%)

Threat Explanation:
gov-db-srv-01 shows 203 connect() calls in 10 seconds vs baseline 7.9/s (+5.4σ). Combined with ALT-001/ALT-002 origin on gov-app-srv-04, this is consistent with an internal sweep launched after initial compromise - the attacker mapping the internal network topology before pivoting. Port sweep pattern matches T1046 tooling seen in CERT-In CI-2024-0089 pre-ransomware reconnaissance phase.

Predicted Next Move: T1021 Remote Services (SSH/RDP brute force or credential replay) against gov-auth-srv within 10-20 minutes.

Immediate Containment:
1. Block outbound from gov-db-srv-01 to gov-auth-srv at firewall
2. Trigger human gate before isolating db-srv-01 (blast radius > 5 critical assets)

V. PHASE 4: ATTACK GRAPH AGENT OUTPUT

```
$ python3 agents/graph_agent.py -origin gov-app-srv-
```

```
[03:13:58] AttackGraphAgent: loading asset graph from
Neo4j...
[03:13:58] Graph: 23 nodes | 41 edges | loaded in 0.18
s

[03:13:58] mark_compromised("gov-app-srv-04")

[03:13:58] predict_next_targets(origin="gov-app-srv
-04", top_k=5)
Traversing up to depth=3 paths... 487 paths
evaluated

PREDICTED NEXT TARGETS:
#1 gov-auth-srv criticality=0.95 hops=2 score
=4.71
#2 gov-db-srv-01 criticality=0.90 hops=1 score
=4.20 [ALERT ALT-003 ACTIVE]
#3 gov-backup-01 criticality=0.80 hops=2 score
=3.84
#4 gov-mail-srv criticality=0.60 hops=2 score
=2.11
#5 gov-web-srv-02 criticality=0.50 hops=1 score
=1.90

[03:13:59] blast_radius(origin="gov-app-srv-04")

BLAST RADIUS REPORT:
Total reachable assets : 8
Critical (crit > 0.7) : 3
-> gov-auth-srv [CRITICAL domain-wide
credential store]
-> gov-db-srv-01 [CRITICAL citizen PII database]
-> gov-backup-01 [HIGH backup & recovery
systems]
Estimated blast : HIGH

ASSET REACHABILITY TREE:
gov-app-srv-04 [COMPROMISED]
|-- gov-web-srv-02 [MEDIUM] port 80/443
internal API
|-- gov-db-srv-01 [HIGH] MySQL 8.0, port
3306
| |-- gov-backup-01 [HIGH] rsync daemon,
port 873
| `-- gov-auth-srv [CRITICAL] LDAP/Kerberos,
port 389/88
`-- gov-mail-srv [MEDIUM] Postfix, port 25

HUMAN GATE TRIGGERED: critical_assets=3 > threshold=2
Automated isolation of gov-db-srv-01 SUSPENDED
pending analyst approval.
Reason: collateral impact on 3 downstream critical
services.
```

VI. PHASE 5: SOAR ORCHESTRATOR AUDIT TRAIL

The orchestrator executes the CRITICAL playbook for ALT-001/ALT-002 and logs every action to an immutable JSONL audit file (SHA-256 signed).

```
orchestrator.py - stdout (incident KS-2026-062201)
```

```
[03:13:14] ResponseOrchestrator: CRITICAL playbook
triggered
host=gov-app-srv-04 | alerts=[ALT-001, ALT
-002]
blast_radius=HIGH | critical_assets=3

[03:13:14] Executing: SNAPSHOT_VM
-> Hypervisor API: POST /vms/gov-app-srv
-04/snapshot
-> Snapshot ID: snap-20260622-031314
[ACK] Forensic snapshot created at 2026-06-22T03:13:14
Z
```

```
[03:13:16] Executing: ISOLATE_HOST
-> SDN controller: PATCH /vlangs/quarantine/
members
-> gov-app-srv-04 moved from VLAN-10 to
VLAN-99 (quarantine)
-> All inbound/outbound blocked except
management interface
[ACK] gov-app-srv-04 isolated in quarantine VLAN-99

[03:13:18] Executing: REVOKE_CREDENTIAL
-> IAM API: DELETE /sessions?host=gov-app-
srv-04
-> Terminated 3 active sessions: uid=[
svc_etl, appuser, root]
-> LDAP bind tokens invalidated for
affected accounts
[ACK] 3 sessions revoked | 3 LDAP tokens invalidated

[03:13:19] Executing: ESCALATE_HUMAN
-> PagerDuty API: POST /incidents (priority
=P1)
-> Incident #PD-20260622-0041 created
-> On-call SOC analyst notified via:
PagerDuty + SMS + Slack #soc-alerts
[ACK] PagerDuty P1 fired | SOC notified

[03:13:20] Executing: BLOCK_IP (ALT-003 lateral
movement source)
-> Firewall API: POST /rules {src:gov-app-
srv-04, dst:gov-db-srv-01,
ports:[3306,873,389,88], action:DROP}
[ACK] Firewall rule ID fw-20260622-0087 injected

[03:14:02] HUMAN GATE: gov-db-srv-01 isolation
SUSPENDED
blast_radius.critical_assets=3 > threshold
=2
Awaiting analyst decision via SOC dashboard
...

[03:17:44] ANALYST DECISION: APPROVE isolation of gov-
db-srv-01
analyst=soc_analyst_01 | ticket=KS
-2026-062201
[03:17:44] Executing: ISOLATE_HOST (gov-db-srv-01)
[ACK] gov-db-srv-01 moved to VLAN-99
```

INCIDENT KS-2026-062201 SUMMARY

```
Time first eBPF alert : 2026-06-22 03:13:11 UTC
Time first host isolated : 2026-06-22 03:13:16 UTC (
MTTD+MTTR: 46s)
Total automated actions : 5
Human gate decisions : 1
Hosts quarantined : 2 (gov-app-srv-04, gov-db-
srv-01)
Sessions revoked : 3
Firewall rules injected : 1
Forensic snapshots : 1
Audit log entries : 7 (SHA-256 signed, tamper-
evident)
```

A. JSONL Audit Log (excerpt)

The audit log below is the immutable record emitted to /var/log/ks_audit.jsonl. Every entry is SHA-256 chained.

```
/var/log/ks_audit.jsonl (last 7 entries)
```

```
{"seq":1,"ts":"2026-06-22T03:13:14Z","action":"
snapshot_vm",
"target":"gov-app-srv-04","result":"snap
-20260622-031314",
"prev_hash":"0000000000","hash":"a3f9d12e..."}
```

```

{"seq":2,"ts":"2026-06-22T03:13:16Z","action":"
  isolate_host",
  "target":"gov-app-srv-04","vlan_from":10,"vlan_to":99,
  "result":"ACK","prev_hash":"a3f9d12e...","hash":"
    b7c2e841..."}

{"seq":3,"ts":"2026-06-22T03:13:18Z","action":"
  revoke_credential",
  "target":"gov-app-srv-04","sessions_terminated":3,
  "uids":["svc_etl","appuser","root"],
  "result":"ACK","prev_hash":"b7c2e841...","hash":"
    c1d4f903..."}

{"seq":4,"ts":"2026-06-22T03:13:19Z","action":"
  escalate_human",
  "target":"soc_analyst_01","pd_incident":"PD
    -20260622-0041",
  "result":"ACK","prev_hash":"c1d4f903...","hash":"
    d8a2b671..."}

{"seq":5,"ts":"2026-06-22T03:13:20Z","action":"block_ip
  ",
  "target":"gov-app-srv-04->gov-db-srv-01","rule_id":"fw
    -20260622-0087",
  "result":"ACK","prev_hash":"d8a2b671...","hash":"
    e5f3c429..."}

{"seq":6,"ts":"2026-06-22T03:14:02Z","action":"
  HUMAN_GATE",
  "target":"gov-db-srv-01","reason":"
    blast_radius_critical_assets=3",
  "result":"SUSPENDED","prev_hash":"e5f3c429...","hash":"
    f0e1d847..."}

{"seq":7,"ts":"2026-06-22T03:17:44Z","action":"
  isolate_host",
  "target":"gov-db-srv-01","analyst":"soc_analyst_01",
  "ticket":"KS-2026-062201","vlan_from":10,"vlan_to":99,
  "result":"ACK","prev_hash":"f0e1d847...","hash":"
    a9b3c512..."}

```

VII. PHASE 6: SOC COPILOT DIALOGUE TRACE

Full multi-turn SOC Copilot session captured during the incident. Claude Sonnet 4.6. Session ID: KS-2026-062201.

SOC Copilot - Full Chat Trace (03:13:25 - 03:19:10 UTC)

```

[03:13:25] SESSION OPENED | analyst=soc_analyst_01
  Context injected: 4 alerts, 2 RAG analyses,
  blast_radius report
  Model: claude-sonnet-4-6 | tokens_budget
    =8192

```

ANALYST > Why was ALT-001 triggered?

COPILLOT > [tokens_in=2341 tokens_out=198 latency=1.3s]

ALT-001 fired at 03:13:11 UTC on gov-app-srv-04.

Root cause: openat() syscall rate hit 847/s vs 7-day baseline of 11.8/s -- a +6.2 sigma deviation captured by eBPF tracepoint sys_enter_openat. Concurrent execve of 'find / -name '*.conf'` (PID 41204, ppid=41203) confirms this was automated file enumeration, not a legitimate process spike.

TTP: T1083 File and Directory Discovery (MITRE ATT&CK, 94% confidence). Matched against APT29 signature in CERT-In CI -2024-0312.

Related: ALT-002 (T1005, same PID) fired 2 seconds later -- read()

burst on the same process confirms active data collection underway.

ANALYST > Are the alerts on both hosts related?

COPILLOT > [tokens_in=2539 tokens_out=231 latency=1.4s]

Yes -- high confidence they are one campaign.

Evidence chain:

1. ALT-001/002: Discovery + Collection on gov-app-srv-04 (03:13:11)
2. ALT-003: connect() burst on gov-db-srv-01 (03:13:55) -- 44 seconds after initial compromise, consistent with automated lateral movement script execution.
3. Attack graph confirms direct edge: gov-app-srv-04 -> gov-db-srv-01 (MySQL port 3306). The timing and TTP chain map to APT29 standard playbook: T1083 -> T1005 -> T1046 -> T1021.

ALT-004 (futex spike on db-srv-01) may indicate a thread-injection attempt or simply lock contention from the port scan -- confidence lower (2.1 sigma, MEDIUM). Monitoring.

ANALYST > What has been done automatically so far?

COPILLOT > [tokens_in=2702 tokens_out=187 latency=1.2s]

5 automated actions completed (all logged in audit trail):

03:13:14	snapshot_vm	gov-app-srv-04	[snap
-20260622-031314]			
03:13:16	isolate_host	gov-app-srv-04	[VLAN-99
quarantine]			
03:13:18	revoke_credential	gov-app-srv-04	[3
sessions killed]			
03:13:19	escalate_human	PagerDuty P1	[PD
-20260622-0041]			
03:13:20	block_ip	app->db firewall	[rule fw
-20260622-0087]			

1 action SUSPENDED (human gate):

03:14:02 isolate_host gov-db-srv-01 [awaiting YOUR approval]

Reason for gate: blast radius analysis shows 3 critical downstream assets (gov-auth-srv, gov-backup-01) that would lose connectivity if db-srv-01 is quarantined without a maintenance window.

ANALYST > What should I do right now? Give me a prioritised list.

COPILLOT > [tokens_in=2891 tokens_out=276 latency=1.6s]

Immediate priorities (next 10 minutes):

- [1] APPROVE isolation of gov-db-srv-01. The firewall block (fw-20260622-0087) limits new connections but the host is not yet quarantined. Risk of attacker pivoting to gov-auth-srv via existing session is HIGH. Use the dashboard

```
gate button or: POST /api/gate/KS-2026-062201/
approve
```

```
[2] CHECK gov-auth-srv for T1078 (Valid Accounts).
Attack graph predicts this as next target (
criticality=0.95).
Look for: unusual LDAP bind attempts, Kerberos
TGT requests from
gov-db-srv-01 IP range, failed login bursts on
port 389/88.
```

```
[3] PULL the forensic snapshot (snap
-20260622-031314).
Memory image of gov-app-srv-04 at time of
compromise is already
saved. Start analysis with: strings +
volatility3 to recover
in-memory credentials before they decay.
```

```
[4] SIGN the CERT-In incident report (IR template
auto-generated).
Mandatory reporting within 6 hours of detection
per CERT-In
guidelines. File: /reports/CERT-In-IR-KS
-2026-062201.pdf
Needs your digital signature before submission.
```

```
[5] MONITOR gov-backup-01 (no alerts yet,
criticality=0.80).
If attacker reaches backup systems, recovery
capability is at risk.
Enable enhanced logging: kernelsentinel watch --
host gov-backup-01
```

```
[03:19:10] SESSION CLOSED | duration=5m45s | turns=4
Total LLM tokens: in=10473 out=892 | cost=
$0.021
```

VIII. END-TO-END TIMELINE

Time (UTC)	Event	Detail
03:12:30	Demo start	Attack trace injected into CICIDS2017 stream
03:13:10	eBPF spike	openat 847/s detected (T+40s from injection)
03:13:11	ALT-001 emitted	+6.2 σ , CRITICAL, T1083
03:13:13	ALT-002 emitted	+8.1 σ , CRITICAL, T1005
03:13:14	Snapshot taken	Forensic VM image, 12.3 GB
03:13:16	Host isolated	gov-app-srv-04 $\rightarrow$ VLAN-99
03:13:18	Creds revoked	3 sessions, 3 LDAP tokens
03:13:19	PagerDuty P1	SOC analyst paged
03:13:20	Firewall rule	app-srv-04 $\rightarrow$ db-srv-01 blocked
03:13:25	SOC Copilot	Analyst opens session
03:13:55	ALT-003 emitted	+5.4 σ , HIGH, T1046
03:13:58	Attack graph	Blast radius: HIGH, 3 critical assets
03:14:02	Human gate	db-srv-01 isolation suspended
03:15:22	ALT-004 emitted	+2.1 σ , MEDIUM, T1055
03:17:44	Gate approved	Analyst approves db-srv-01 isolation
03:17:44	db-srv-01 isolated	Second host quarantined
03:19:10	Copilot closed	4-turn session, CERT-In report ready
03:13:11–03:13:16	MTTD+MTTR	46 seconds (auto actions)
Baseline SOC	MTTD	21 days (industry avg.)

TABLE II: End-to-end incident timeline

IX. KEY RESULT: DETECTION LEAD TIME

Metric	Baseline SOC	KernelSentinel
MTTD	21 days	43 seconds
MTTR (auto actions)	4 hours	46 seconds
Host isolation time	manual, hours	5 seconds
Forensic snapshot	post-mortem	3 seconds
Cred revocation	manual, 30 min	2 seconds
Audit trail	manual log	100%, SHA-256 chained
LLM analysis latency	N/A	1.2–1.6s per query
False positive (demo)	–	0 / 4 alerts
Human gate decisions	all manual	1 / 6 actions

X. CONCLUSION

The KernelSentinel demo run on an injected APT29-pattern trace demonstrates a complete detection-to-containment pipeline executing in under 50 seconds, compared to an industry-average MTTD of 21 days for signature-based SOC operations. The eBPF telemetry layer provides kernel-level behavioural visibility with zero overhead on untraced processes – a capability validated by thesis benchmark data (IIT Jammu, 2024–25). The human escalation gate correctly fired when blast radius exceeded the safety threshold, preserving analyst control over high-impact decisions while automating the 5 actions where speed is critical. All actions are immutably logged for legal admissibility. Full source code and datasets available at demo time.

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